



Envision Energy PCS PSCAD Model User Manual

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1 Introduction

The Envision PCS model is a simulation model developed based on PSCAD/EMTDC simulation software, mainly used to support the grid-connection simulation testing.

This document focuses on the usage and configuration of the PCS model. For the usage of PPC model, please refer to PPC model user manual.

2 Model Overview

The PCS and PPC (power plant controller) are modelled separately in PSCAD, and are provided together as a ready simulation case when both models are required, as model main page shown in Figure 1 for example.

The cables and main transformer are set up for testing purpose only and user should configure them as actual plant BoP design accordingly. The “Envision POC Meter” module is for observation for reference, user could delete or replace them as required.

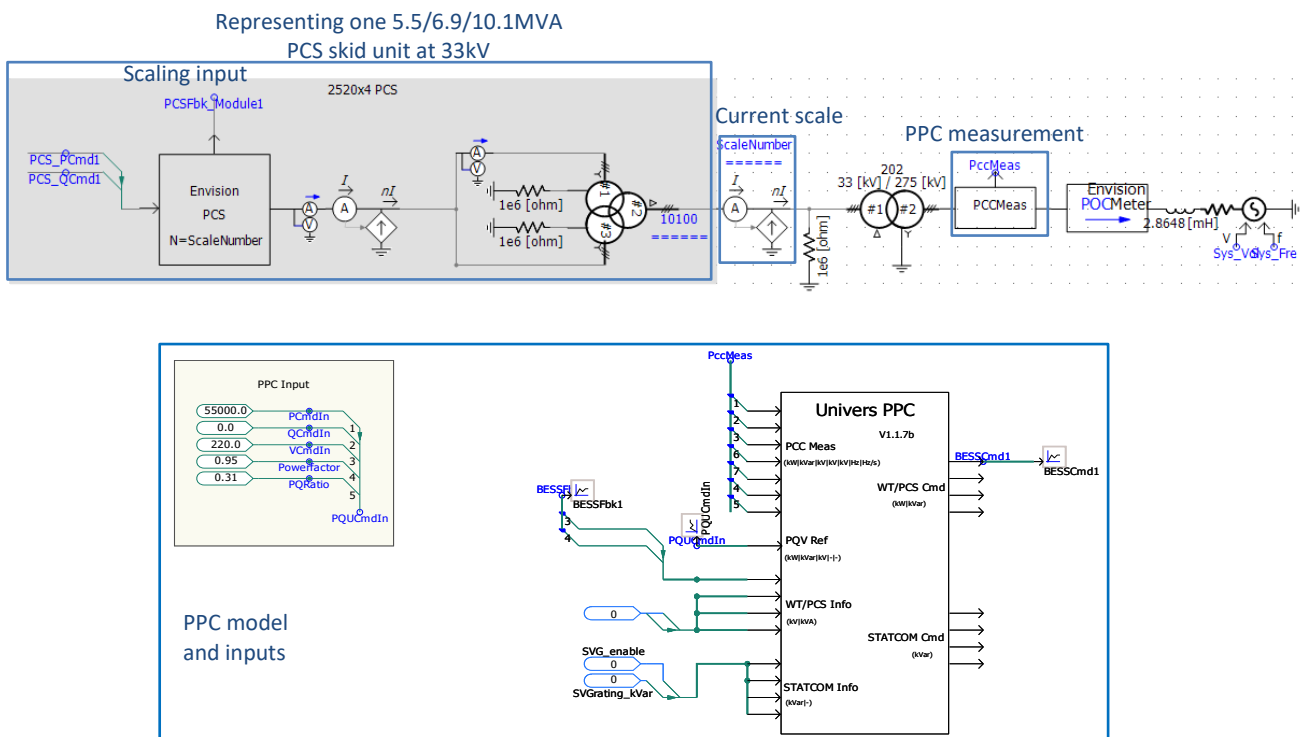


Figure 1 Model main page illustration (model name may differ for different versions)

Each PCS electrical topology is shown in Figure 2, featuring two converters connected in parallel.

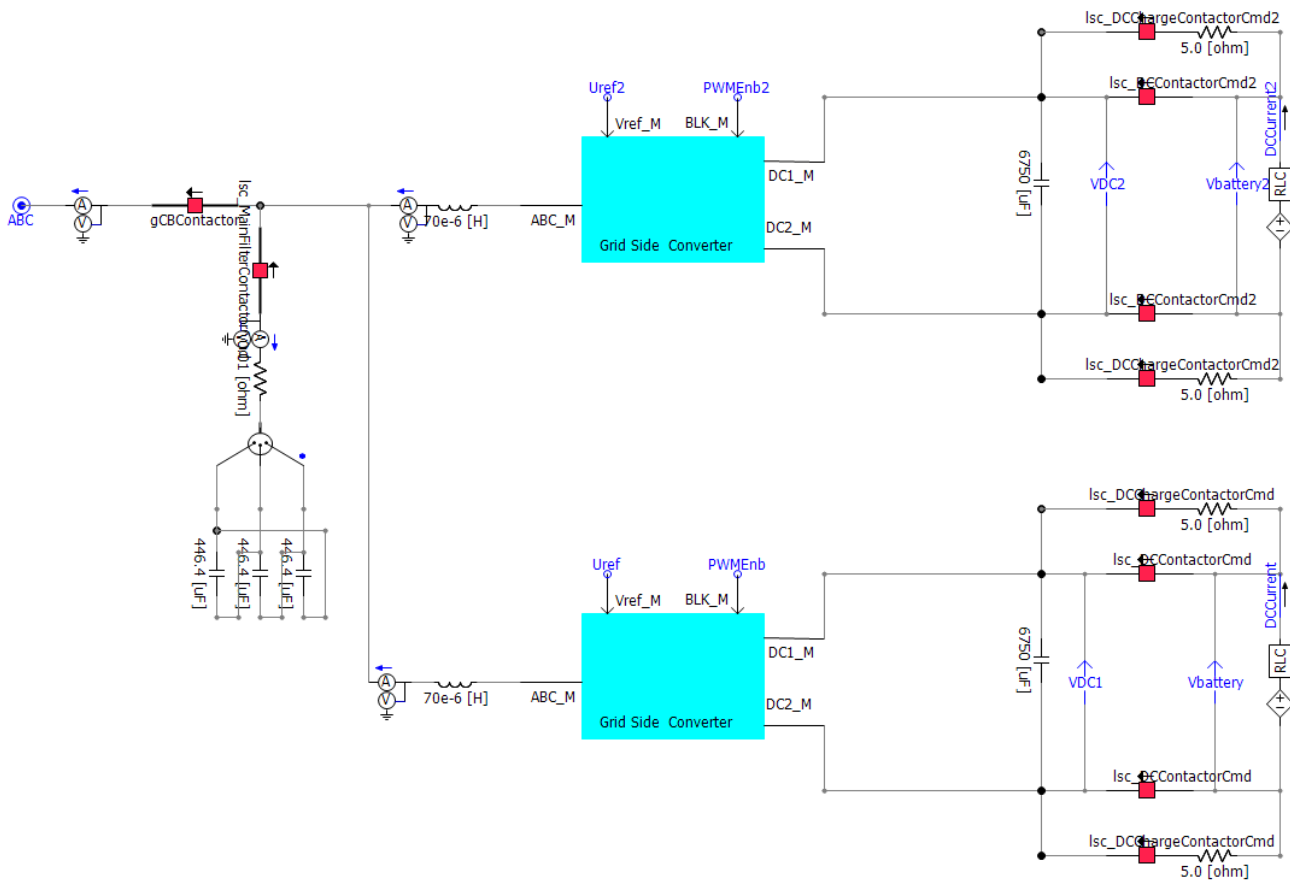


Figure 2 PCS electrical connection

The PCS PSCAD model simplifies the DC side controller since the battery can be taken as stable current source. The other functionality/modules are represented with real control source code of the PCS, thus all of features such as saturations, time-delay, non-linearity, dead-band, wind-up are well included.

PCS operates in active power and reactive power control mode, that is, the PCS responds to the external (PPC or local) reactive and active power commands.

PPC can operate in reactive power control mode, power factor control mode, direct voltage control mode, or voltage droop control mode; active power control mode and frequency response control mode are also provided.

3 Model Setup and Dependencies

The compressed file obtained by users should contain two folders named "dll" and "Interface" respectively, with the PSCAD model as "pscxd" format. The 'dll' folder includes the .dll file which is the real Wind Turbine source code compiled model. When configuring the simulation project, user needs to make sure that the "dll" and "Interface" folders are placed in the same path as the PSCAD project file.

To run the PCS with PPC PSCAD model correctly, it is recommended to use PSCAD V5.0 and intel Fortran Compiler XE 15 and above. The PSCAD model file name will indicate if it works with 32bit or 64bit compiler.

The 'interface' folder includes necessary .obj files for the model to operate, user needs to make sure all the .obj files are in the 'resource' folder in PSCAD, as example shown in the figure below.

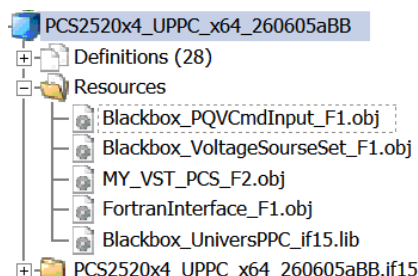


Figure 3 Link object files in PSCAD

The Intel Fortran compiler should also be configured correctly before running the simulation, otherwise the object files cannot be linked and the dll files cannot be loaded appropriately, leading to failure in running the model.

The simulation time step works at anywhere between rang 1us ~ 200us, and 50us recommended, and the model is represented with current source.

4 Configuration and parameter settings

4.1 PCS model scaling information

The current scaling component at the output of the PCS model reads scaling setting that can be configured by user, as shown in Figure 4.

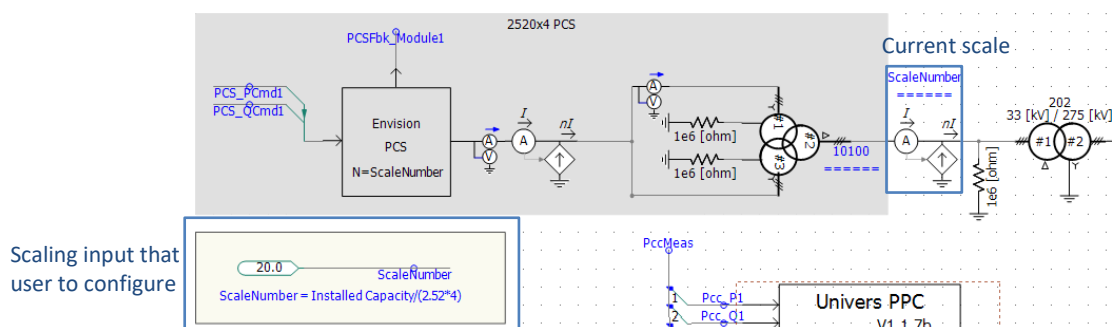


Figure 4 PCS scale number setting

4.2 PCS model input and output

The PCS models receives active power command and reactive power command form PPC model via multi-dimensional signal 'BESSCmdx' and provide required feedback variables to PPC via 'BESSFBKx'. With PPC model implemented, the overall model receives PPC command inputs to regulate the plant active power and reactive power at plant Point of Connection.

The settings in the PCS model menu are for Envision R&D to configure, user please do not change those settings.

4.3 PCS Protection Setting

PCS PSCAD protection consists of low voltage, high voltage, low frequency and high frequency protection functions. By double clicking on the “Envision Volt & Freq Protection”, user could read the protection settings and configure if required, as the menu shown in Figure below.

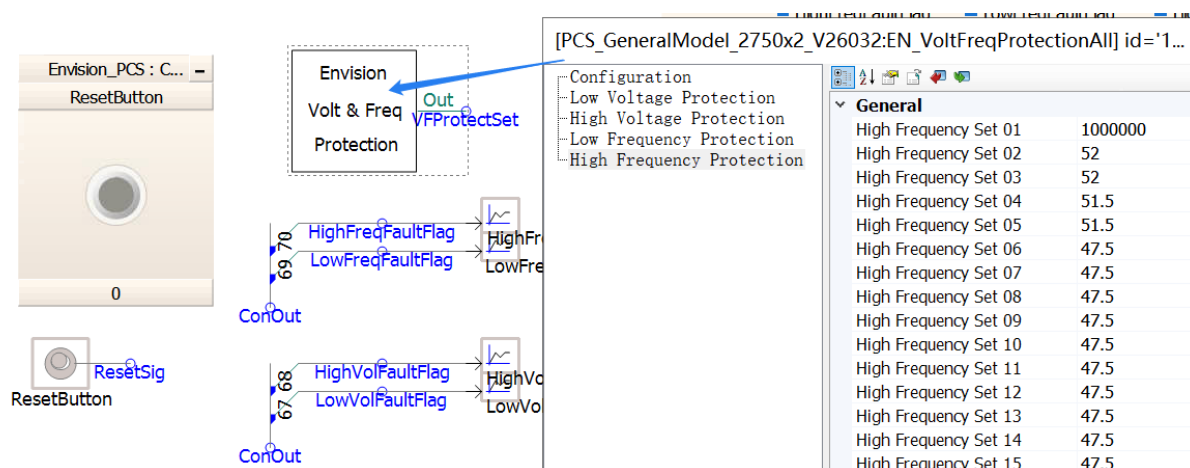


Figure 5 protection settings interface

All protection functions are configured using a point-based definition method. Users specify discrete voltage–time (V–t) or frequency–time (f–t) points, and the protection characteristic curve is generated by linear interpolation between these points. Voltage values are defined in per unit (p.u.), frequency values in hertz (Hz), and time durations in seconds (s).

Each protection type generates an independent fault signal, including low-voltage, high-voltage, low-frequency, and high-frequency faults. When a protection criterion is met, the corresponding fault signal is activated. If a fault signal outputs a fault code with value x , it indicates that the voltage or frequency protection level defined between Set x and Set $(x+1)$ has been triggered for the corresponding voltage or frequency range.

A trip may result in multiple protection functions being triggered in sequence. Therefore, the first activated protection flag should be considered as the primary cause of PCS trip.

As an example illustrated in the figure below, the high-frequency protection fault is triggered with a fault code of 1. This indicates that the high frequency range defined by Set 1 to Set 2 has been activated, which is consistent with the configured frequency protection settings.

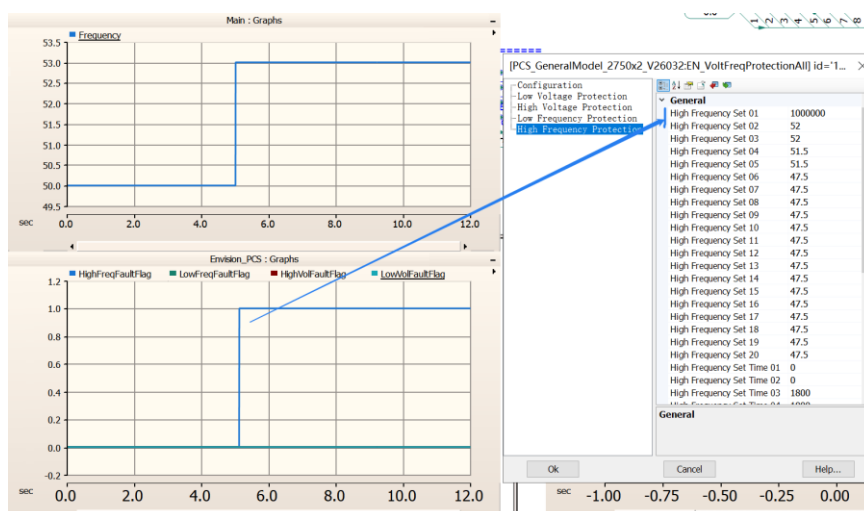


Figure 6 fault code illustration example